A 2-COMMUTATIVE SQUARE INDUCES A FUNCTOR BETWEEN FINITE HILBERT STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Closing routine-check omission retained verbatim and flagged for the reviewer.

Allowed standard prerequisites (editorial).

- Horizontal composition. categories-definition-horizontal-composition (cross-chapter, named). Fixes the whiskering notation gamma * id_H * alpha'.

Source reference. Original reference 003E: focused-seeds/geometry/S9-crit-map-hilbert/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05WD: focused-seeds/geometry/S9-crit-map-hilbert/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a* 2*-commutative diagram*

$$\begin{array}{ccc} \mathcal{X}' & \xrightarrow{G} & \mathcal{X} \\ {\scriptstyle F'}\downarrow & & \downarrow{\scriptstyle F} \\ \mathcal{Y}' & \xrightarrow{H} & \mathcal{Y} \end{array}$$

of stacks in groupoids over $(Sch/S)_{fppf}$ *with a given* 2*-isomorphism* $\gamma : H \circ F' \to F \circ G$*. In this situation we obtain a canonical* 1*-morphism* $\mathcal{H}_d(\mathcal{X}'/\mathcal{Y}') \to \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$*. This morphism is compatible with the forgetful* 1*-morphisms of Examples of Stacks, Equation (05WD).*

Proof. We map the object (U, Z, y', x', α') to the object $(U, Z, H(y'), G(x'), \gamma \star \mathrm{id}_H \star \alpha')$ where $\star$ denotes horizontal composition of 2-morphisms, see Categories, Definition 003E. To a morphism $(f, g, b, a) : (U_1, Z_1, y'_1, x'_1, \alpha'_1) \to (U_2, Z_2, y'_2, x'_2, \alpha'_2)$ we assign $(f, g, H(b), G(a))$. We omit the verification that this defines a functor between categories over $(Sch/S)_{fppf}$. □

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